ADITYA SINGH

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PROFESSIONAL SUMMARY

Electronics & Communication Engineering graduate with real-world embedded firmware experience as Flight Computer Lead at an aerospace startup. Skilled in bare-metal and Linux-based embedded C firmware development, real-time sensor interfacing, interrupt-driven control systems, and hardware-software integration. Proficient in communication protocols (I2C, SPI, UART), Python scripting, and low-power system design. Passionate about building reliable, efficient embedded software at the hardware-software boundary.

EXPERIENCE

Flight Computer Lead | Aishyam Aerospace | *2022 – 2023*

Led embedded firmware development for the flight computer of an aerospace-grade system at an early-stage startup.

- Owned end-to-end flight computer firmware in Embedded C on a Raspberry Pi-based Linux platform, covering boot sequencing, peripheral initialization, and runtime control loops
- Developed and integrated sensor driver firmware for flight-critical real-time data acquisition under deterministic timing constraints
- Designed fault-aware firmware with error detection and safe-state handling logic to ensure reliable system behavior under failure conditions
- Collaborated with hardware and avionics teams to define firmware-hardware interfaces from performance, compatibility, and integration perspectives
- Made independent firmware architecture decisions as sole firmware owner, driving the system from bring-up through functional validation
- Gained practical experience with embedded Linux system programming, hardware peripheral access, and process management

PROJECTS

Solar-Powered Remote Water Tank Monitoring & Control System | Academic Project | *2023*

- Developed autonomous bare-metal firmware in Embedded C for real-time tank level monitoring and automatic pump ON/OFF control using sensor-based decision logic
- Implemented fail-safe protection mechanisms including dry-run detection and overflow prevention for safe motor operation
- Designed timer and interrupt-based firmware architecture for responsive, deterministic control without an OS
- Optimized firmware for solar-powered, low-power continuous operation using duty-cycling and sleep modes
- Enabled remote monitoring of system status via ESP32 wireless communication and developed a hardware abstraction layer for sensors, relays, and actuators

Smart Autonomous Farming System | Academic Project | *2023*

- Built a multi-sensor embedded system for soil monitoring, motor control, intrusion detection, and remote monitoring using Embedded C on Arduino/ESP32
- Designed modular firmware architecture with independent sensor driver modules and hardware abstraction for scalable system design
- Integrated wireless communication for remote monitoring and engineered low-power solar operation with power management firmware

Gesture-Controlled LED System Using Computer Vision | Academic Project | *2022*

- Developed a real-time Python/OpenCV image processing pipeline for hand gesture recognition using frame preprocessing, contour detection, and thresholding
- Implemented UART serial communication between the host Python application and microcontroller firmware to toggle LED states based on detected gestures
- Integrated computer vision and embedded control, enabling touch-free device operation across two hardware domains

TECHNICAL SKILLS

Programming Languages: Embedded C, Python, C++ (learning), Assembly (fundamentals)
Platforms & OS: Raspberry Pi (Embedded Linux), Arduino (ATmega), ESP32
Communication Protocols: I2C, SPI, UART, CAN (learning)
Firmware Concepts: Bare-metal development, embedded Linux firmware, interrupt-driven design, timer/PWM control, low-power optimization, sensor driver development, hardware abstraction, fault-safe state handling
Tools & Environment: Arduino IDE, Serial Monitor, Git, OpenCV
Core CS: Data Structures & Algorithms, OS, Computer Architecture, Computer Networks
Hardware: Sensors (ultrasonic, soil moisture, PIR, IMU), motors, relays, signal conditioning

EDUCATION

B.Tech in Electronics & Communication Engineering | Academy of Technology | *2020 – 2024*
CGPA: 8.63/10 | Relevant coursework: Digital Logic Design, Embedded Systems, Computer Organization, Microprocessors & Microcontrollers, Signal Processing

Higher Secondary Education (Science) | Abbot Shishu Hall | *2018 – 2020*
Percentage: 85%

CERTIFICATIONS

- The Arduino Platform and C Programming — University of California, Irvine (Coursera)
- Interfacing with the Arduino — University of California, Irvine (Coursera)
- Introduction to IoT and Embedded Systems — University of California, Irvine (Coursera)
- The Raspberry Pi Platform and Python Programming — University of California, Irvine (Coursera)

LANGUAGES

English (Professional) | Hindi (Native) | Bengali (Native)